CRT State Engine - Professional Operational Manual

Version 1.0 | For Intermediate to Advanced Algorithmic Traders

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System Overview

What is the CRT State Engine?

The **Candle Range Theory (CRT) State Engine** is a comprehensive institutional trading system that programmatically detects and trades high-probability market manipulation patterns used by institutional traders. The system combines real-time market analysis, session-based state tracking, and precise entry signal generation.

Core Methodology

The system is built on three fundamental pillars:

1. **Draw on Liquidity (DOL) Analysis** - Determines institutional bias and target zones
2. **Session State Machine** - Tracks market behavior across trading sessions
3. **Multi-Model CRT Detection** - Identifies 1AM, 5AM, and 9AM institutional patterns

Key Features

- **Real-time DOL detection** with automatic bias calculation
- **Session behavior classification** (Consolidation, Manipulation, Expansion)
- **Three CRT models** covering different institutional time windows
- **Order Block and FVG detection** for precise entry timing
- **Comprehensive alert system** with detailed market context
- **Risk management integration** with automatic R:R calculations

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Installation & Setup

Prerequisites

- TradingView Pro account (for real-time data and alerts)
- Access to 4-hour timeframe charts
- Basic understanding of institutional trading concepts

Installation Steps

1. **Copy the Pine Script code** from the provided file
2. **Open TradingView** and navigate to Chart → Pine Script Editor
3. **Paste the code** and click "Add to Chart"
4. **Configure initial settings** according to your trading preferences
5. **Test on historical data** to verify proper installation

Initial Configuration

Essential Settings:

```
Core System:
✅ Enable DOL Analysis Engine
✅ Enable Session State Machine
✅ Enable Live Entry Signals
❌ Debug Mode (disable for live trading)

DOL Analysis:
- DOL Pivot Strength: 10 (default, increase for less sensitive detection)

CRT Models:
✅ 1AM CRT Model (London manipulation)
✅ 5AM CRT Model (London lunch transition)
✅ 9AM CRT Model (NY session commitment)
```

Recommended Visual Settings:

```
Visuals:
✅ Show DOL Lines
✅ Show Session Boxes
✅ Show Key Levels
✅ Show Entry Signals
```

Core Components

1. DOL (Draw on Liquidity) Analysis Engine

Purpose: Determines the institutional directional bias by identifying the nearest significant liquidity pool.

How it Works:

- Scans for major swing highs and lows using configurable pivot strength

- Calculates distance from current price to each liquidity zone
- Determines bias based on "path of least resistance" principle
- Updates automatically as new swing points are confirmed

Dashboard Display:

```
DOL Bias: BULLISH → 1.0980
```

- **BULLISH**: Price likely to move toward swing high target
- **BEARISH**: Price likely to move toward swing low target
- **Target Price**: Specific level where liquidity resides

2. Session State Machine

Purpose: Classifies market behavior across different trading sessions to understand institutional intent.

Session Windows (GMT):

- **CBDR**: 17:00-21:00 (Close, Reversal, Discount, Premium)
- **ASIA**: 21:00-01:00 (Asian session activity)
- **LONDON**: 01:00-05:00 (London session manipulation)
- **LUNCH**: 05:00-09:00 (London lunch, NY preparation)
- **NY**: 09:00-13:00 (New York session expansion)

Behavior Classification:

- **CONSOLIDATION**: Range-bound, accumulation phase
- **MANIPULATION**: False breakouts, liquidity sweeps
- **EXPANSION**: Directional movement, trend continuation

3. CRT Pattern Detection

1AM CRT Model

Target Window: London session (01:00-05:00 GMT) **Patterns:**

- **Normal Protraction**: CBDR consolidation → Asia consolidation → London manipulation
- **Delayed Protraction**: CBDR consolidation → Asia manipulation → London expansion

States:

- **A_FORMING**: Accumulation phase beginning
- **M_LIVE**: Manipulation in progress
- **D_READY**: Distribution phase, entry zone active

5AM CRT Model

Target Window: London lunch transition (05:00-09:00 GMT) **Patterns:**

- **London Lunch Low**: Market establishes daily low during lunch
- **NY Continuation**: Continuation of London direction into NY
- **NY Reversal**: Reversal of prior session trend

9AM CRT Model

Target Window: NY session opening (09:00-13:00 GMT) **Patterns:**

- **NY Continuation**: DOL not reached, London made HOD/LOD
- **NY Reversal**: DOL reached, protected session highs/lows

Understanding the Dashboard

Main Dashboard Layout

```
┌──────────────────────────────────────────┐
│        CRT STATE ENGINE                  │
├──────────────────┬──────────┬────────────┤
│ DOL Bias         │ BULLISH  │ 1.0980     │
│ Session          │ LONDON   │ MANIP      │
│ 1AM CRT          │ D_READY  │ NORMAL     │
│ 5AM CRT          │ INACTIVE │            │
│ 9AM CRT          │ INACTIVE │            │
│ Entry Signal     │ NONE     │ -          │
│ Key Levels       │ OB BULL  │ FVG BEAR   │
└──────────────────┴──────────┴────────────┘
```

Dashboard Interpretation

DOL Status Row

- **Left**: Always shows "DOL Bias"
- **Middle**: Current bias direction (BULLISH/BEARISH/NEUTRAL)
- **Right**: Target price level
- **Color**: Green (bullish), Red (bearish), Gray (neutral)

Session Status Row

- **Left**: Always shows "Session"
- **Middle**: Current session name
- **Right**: Session behavior type
- **Color**: Gray (consolidation), Orange (manipulation), Blue (expansion)

CRT Model Rows

- **Left**: Model name (1AM/5AM/9AM CRT)
- **Middle**: Current state (INACTIVE/A_FORMING/M_LIVE/D_READY)
- **Right**: Active profile type (when applicable)

- **Color**: Green (D_READY), Gray (other states)

Entry Signal Row

- **Left**: Always shows "Entry Signal"
- **Middle**: Signal type (NONE/BUY/SELL)
- **Right**: Entry price (when signal active)
- **Color**: Green (BUY), Red (SELL), Gray (NONE)

Key Levels Row

- **Left**: Always shows "Key Levels"
- **Middle**: Order Block status
- **Right**: Fair Value Gap status
- **Color**: Orange (OB detected), Purple (FVG detected), Gray (none)

Trading Methodology

The Complete Trading Process

Phase 1: Market Context Analysis (H4 Timeframe)

Step 1: DOL Assessment

✅ Identify current DOL bias and target
✅ Confirm bias aligns with your directional assumption
✅ Note target price for profit-taking considerations

Step 2: Session Analysis

✅ Determine current session and behavior
✅ Confirm session aligns with trading model
✅ Wait for appropriate session transitions

Step 3: CRT Model Selection

- ✅ Choose primary model based on time availability
- ✅ Enable relevant model in settings
- ✅ Monitor state progression (A → M → D)

Phase 2: Pattern Recognition

1AM CRT Trading:

- **Best for:** London session traders (01:00-09:00 GMT)
- **Entry window:** 02:00-03:00 GMT during D_READY state
- **Profile consideration:** Normal vs Delayed Protraction affects risk/reward

5AM CRT Trading:

- **Best for:** London lunch and NY open traders
- **Entry windows:** 06:00-07:00 GMT (lunch) or 07:00-08:30 GMT (NY prep)
- **Profile consideration:** Three distinct scenarios require different approaches

9AM CRT Trading:

- **Best for:** NY session traders
- **Entry windows:** 09:00-10:00 GMT or 09:30-10:30 GMT
- **Profile consideration:** Continuation vs Reversal based on DOL status

Phase 3: Entry Execution

Entry Conditions (All must align):

- ✅ CRT model in D_READY state
- ✅ DOL bias matches intended direction
- ✅ Within appropriate time window
- ✅ Order Block detected in favor of bias
- ✅ Session behavior supports the move

Entry Methodology:

1. **Monitor dashboard** for D_READY state activation
2. **Switch to M15 timeframe** for precise entry timing
3. **Wait for Order Block formation** in direction of bias
4. **Enter on M15 confirmation** above/below OB levels
5. **Set stops below/above OB** with auto-calculated R:R

Phase 4: Risk Management

Stop Loss Placement:

- **1AM CRT**: Below/above manipulation candle extremes
- **5AM CRT**: Beyond session highs/lows
- **9AM CRT**: Outside 8AM candle range (for H1 version)

Take Profit Targets:

- **Primary**: 1:2 Risk/Reward (default)
- **Extended**: 1:3 Risk/Reward for strong setups
- **Dynamic**: Move to breakeven after 1:1 achieved

Alert System & Interpretation

Alert Priority Levels

🔥 CRITICAL (Immediate Action Required)

```
🚀 BUY SIGNAL GENERATED
🚀 SELL SIGNAL GENERATED
🔥 ENTRY ZONE ACTIVE (D_READY states)
```

⚡ HIGH (Prepare for Action)

🎯 DOL BIAS CHANGED
📅 NEW SESSION STARTED
⚡ SESSION BEHAVIOR CHANGE

📊 MEDIUM (Monitor Closely)

📊 NEW ORDER BLOCK DETECTED
🔄 CRT STATE CHANGE

Alert Message Structure

Every alert contains comprehensive market context:

🚨 CRT STATE ENGINE ALERT 🚨

📊 EURUSD | 240 | 2025-08-13 09:00:00
💰 Price: 1.0955

🎯 MARKET ANALYSIS:
- DOL Bias: BEARISH → 1.0920
- Session: NY (MANIPULATION)
- Active Profile: NORMAL_PROTRACTION

🔄 CRT MODEL STATUS:
- 1AM CRT: D_READY
- 5AM CRT: INACTIVE
- 9AM CRT: INACTIVE

📈 KEY LEVELS:
- Order Block: BEARISH Active
- FVG: None

⚡ ALERT: 🔥 ENTRY ZONE ACTIVE 🔥
📋 Details: 1AM CRT ready for distribution

🤖 CRT State Engine v1.0

Alert Interpretation Guide

DOL Bias Change Alert

When: Swing point analysis determines new institutional target **Action:** Reassess directional bias, adjust trading plan **Significance:** Fundamental shift in market structure

Session Transition Alert

When: New trading session begins (e.g., London → Lunch) **Action:** Prepare for behavior change, monitor for patterns **Significance:** Different sessions have

distinct characteristics

CRT State Change Alert

When: Pattern progresses through A → M → D states **Action:** Increase monitoring intensity, prepare for entries **Significance:** Pattern development in real-time

Entry Zone Active Alert

When: Any CRT model reaches D_READY state **Action:** Switch to execution timeframe, monitor for signals **Significance:** High-probability entry window opening

Entry Signal Alert

When: All conditions align for trade execution **Action:** Execute trade immediately or within 15 minutes **Significance:** Highest conviction trade setup

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Operational Workflow

Daily Routine

Pre-Market Setup (30 minutes before target session)

1. **Check DOL status** - Confirm current bias and targets
2. **Review overnight session behavior** - Understand market context
3. **Enable relevant CRT models** - Based on your trading schedule
4. **Set up alerts** - Ensure all notifications are active
5. **Prepare trading platform** - Ready for quick execution

During Trading Hours

1. **Monitor dashboard continuously** - Watch for state changes
2. **Respond to alerts promptly** - Critical signals require immediate attention

3. **Document decisions** - Keep trading journal with reasoning
4. **Manage open positions** - Follow risk management protocols
5. **Update market analysis** - Adjust strategy based on new information

Post-Session Review (15 minutes after session close)

1. **Review alert history** - Analyze signal accuracy
2. **Document session behavior** - Build pattern recognition
3. **Assess performance** - Compare actual vs expected outcomes
4. **Plan next session** - Identify opportunities and risks
5. **Update settings** - Optimize based on market conditions

Weekly Optimization

Performance Analysis

- **Track alert accuracy** - Hit rate of DOL targets
- **Measure signal quality** - Entry signal win/loss ratio
- **Analyze session classification** - Behavior prediction accuracy
- **Review risk/reward** - Actual vs theoretical R:R ratios

System Tuning

- **Adjust pivot strength** - Based on market volatility
- **Optimize time windows** - Refine entry timing
- **Update alert thresholds** - Reduce noise, increase signal
- **Refine risk parameters** - Adapt to changing market conditions

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Risk Management

Position Sizing

Conservative Approach (Recommended for beginners)

- **Risk per trade:** 0.5-1% of account
- **Maximum concurrent positions:** 2-3
- **Maximum daily risk:** 2% of account

Aggressive Approach (For experienced traders)

- **Risk per trade:** 1-2% of account
- **Maximum concurrent positions:** 3-5
- **Maximum daily risk:** 5% of account

Stop Loss Management

Static Stops

- **1AM CRT**: 20-30 pips beyond manipulation extremes
- **5AM CRT**: 25-40 pips beyond session boundaries
- **9AM CRT**: 15-25 pips beyond pattern invalidation

Dynamic Stops

- **Breakeven move**: After 1:1 R:R achieved
- **Trailing stops**: Follow favorable price action
- **Time-based exits**: Close if setup doesn't develop within 2-4 hours

Risk Scenarios

High-Risk Conditions (Reduce position size or avoid)

- **Major news events** within 2 hours of entry window
- **Low liquidity sessions** (late Friday, holidays)
- **Conflicting DOL signals** (recent bias changes)
- **Multiple failed signals** in same session

Optimal Risk Conditions (Standard position size)

- **Clear DOL bias** maintained for 24+ hours
- **Clean session transitions** with expected behavior
- **Confluence of multiple models** agreeing on direction
- **Strong Order Block formations** supporting bias

Performance Optimization

System Tuning Parameters

DOL Pivot Strength

- **Low volatility markets**: Increase to 12-15 (fewer, stronger signals)
- **High volatility markets**: Decrease to 7-10 (more responsive signals)
- **Range-bound markets**: Increase to 15-20 (avoid noise)

Time Window Adjustments

- **Fast-moving markets**: Narrow windows by 15-30 minutes
- **Slow-moving markets**: Expand windows by 15-30 minutes
- **Holiday sessions**: Expand windows, reduce position size

Alert Sensitivity

- **High-frequency trading**: Enable all alerts, immediate notifications
- **Swing trading**: Focus on D_READY and entry signals only
- **Position trading**: DOL changes and major session transitions

Market Condition Adaptations

Trending Markets

- **Favor continuation models** (9AM NY Continuation)
- **Increase take profits** to 1:3 or 1:4 R:R
- **Trail stops more aggressively**

- **Focus on breakout Order Blocks**

Range-Bound Markets

- **Favor reversal models** (9AM NY Reversal)
- **Take profits earlier** at 1:2 R:R
- **Use tighter stops**
- **Focus on rejection Order Blocks**

High-Impact News Days

- **Avoid trading 2 hours before/after major releases**
- **If already in position**, move to breakeven before news
- **Wait for new DOL establishment** after news impact
- **Reduce position sizes** by 50%

Troubleshooting

Common Issues and Solutions

Dashboard Not Updating

Symptoms: Static values, no state changes **Causes:**

- Timeframe not set to 4H
- Real-time data feed interrupted
- Browser/platform memory issues

Solutions:

1. Refresh chart and re-add indicator
2. Verify timeframe is exactly "240" (4 hours)
3. Check TradingView connection status
4. Clear browser cache if using web platform

Missing Alerts

Symptoms: Expected alerts not firing **Causes:**

- Alert conditions not properly configured
- TradingView alert limits reached
- Notification permissions disabled

Solutions:

1. Check TradingView alert list for active alerts
2. Verify alert conditions match indicator states
3. Test with one alert first, then add others
4. Ensure sufficient alert quota in account

Incorrect DOL Detection

Symptoms: DOL targets seem unrealistic or change too frequently **Causes:**

- Pivot strength too low for market conditions
- Insufficient historical data
- Major market event disrupted swing analysis

Solutions:

1. Increase pivot strength to 12-15
2. Allow 2-3 days for swing point establishment
3. Manually verify swing points align with visual analysis
4. Reset during major market structure changes

Entry Signals Not Generating

Symptoms: D_READY state active but no entry signals **Causes:**

- No Order Block detected in bias direction

- Outside entry time windows
- Risk/reward parameters too strict

Solutions:

1. Switch to M15 timeframe to verify Order Block formation
2. Confirm current time is within model's entry window
3. Check if DOL bias matches expected entry direction
4. Manually verify Order Block criteria on chart

Performance Issues

Slow Chart Loading

Solutions:

- Reduce history days limit to 3-5 days
- Decrease max instances per pattern to 50-100
- Disable debug mode during live trading
- Use shorter lookback periods for calculations

Memory Warnings

Solutions:

- Restart TradingView platform
- Reduce number of active models (use 1-2 instead of all 3)
- Clear browser cache and cookies
- Upgrade to TradingView Pro+ if using basic account

Advanced Features

Multi-Symbol Analysis

Setup Process

1. **Create separate chart windows** f